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# Vacuum Cleaner Agent

class Environment:
    def __init__(self):
        self.locations = {}

    # Take input for environment
    for location in ['A', 'B']:
        while True:
            status = input(
                f"Enter status of Location {location} (Clean/Dirty): "
            ).strip().capitalize()

            if status in ['Clean', 'Dirty']:
                self.locations[location] = status
                break
            else:
                print("Invalid input! Enter Clean or Dirty.")

    # Take initial vacuum cleaner position
    while True:
        position = input(
            "Enter initial vacuum cleaner position (A/B): "
        ).strip().upper()

        if position in ['A', 'B']:
            self.agent_location = position
            break
        else:
            print("Invalid position! Enter A or B.")

class VacuumCleanerAgent:
    def __init__(self, environment):
        self.environment = environment

    def clean(self):
        location = self.environment.agent_location
        status = self.environment.locations[location]

        print(f"\nVacuum cleaner is at Location {location}")
        print(f"Status: {status}")

        if status == "Dirty":
            print("Action: SUCK")
            self.environment.locations[location] = "Clean"
            print(f"Location {location} is now Clean.")

        elif location == "A":
            print("Action: MOVE RIGHT")
            self.environment.agent_location = "B"

        else:
            print("Action: MOVE LEFT")
            self.environment.agent_location = "A"

# Create environment
environment = Environment()

# Create vacuum cleaner agent
agent = VacuumCleanerAgent(environment)

print("\nInitial Environment:")
print(environment.locations)
print("Initial Position:", environment.agent_location)

# Run agent
for i in range(4):
    agent.clean()

print("\nFinal Environment:")
print(environment.locations)

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Enter status of Location A (Clean/Dirty): Dirty
Enter status of Location B (Clean/Dirty): Dirty
Enter initial vacuum cleaner position (A/B): A
```

```
Initial Environment:
{'A': 'Dirty', 'B': 'Dirty'}
Initial Position: A
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```
Vacuum cleaner is at Location A
Status: Dirty
Action: SUCK
Location A is now Clean.
```

```
Vacuum cleaner is at Location A
Status: Clean
Action: MOVE RIGHT
```

```
Vacuum cleaner is at Location B
Status: Dirty
Action: SUCK
Location B is now Clean.
```

```
Vacuum cleaner is at Location B
Status: Clean
Action: MOVE LEFT
```

```
Final Environment:
{'A': 'Clean', 'B': 'Clean'}
```